

# Potential Talents - Technical Report

Candidate ranking, transparent relevance scoring, and recruiter feedback

|                                       |                                      |                                          |                                      |
|---------------------------------------|--------------------------------------|------------------------------------------|--------------------------------------|
| <b>0.948</b><br>NDCG@10 - Aspiring HR | <b>0.935</b><br>NDCG@10 - Seeking HR | <b>100%</b><br>Token occurrence coverage | <b>52</b><br>Unique titles evaluated |
|---------------------------------------|--------------------------------------|------------------------------------------|--------------------------------------|

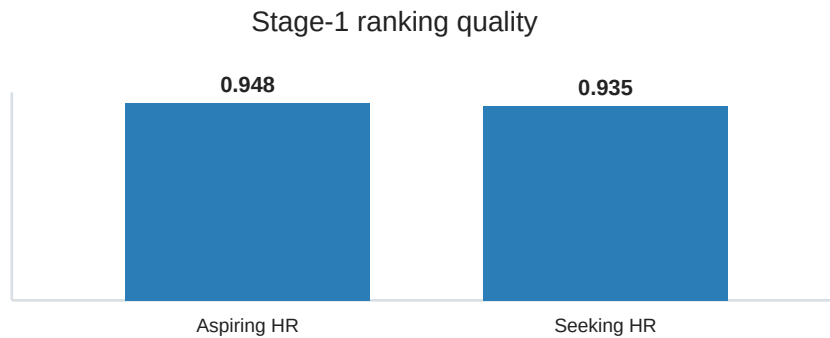
## Executive Summary

The workflow combines duplicate-aware evaluation, a transparent relevance benchmark, semantic candidate ranking, and recruiter-driven reranking. The verified run achieved NDCG@10 of **0.948** and **0.935**, with complete vocabulary coverage.

## Project Objectives

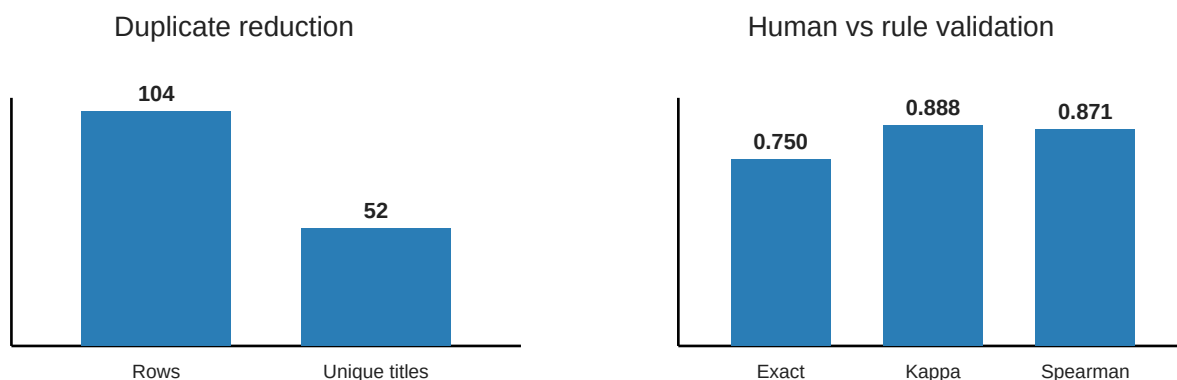
**Overall objective:** Develop a transparent and adaptable candidate-ranking system that helps recruiters identify the most relevant candidates faster and more consistently.

- 1. Data quality.** Remove duplicate titles and analyse vocabulary to prevent bias and support reliable candidate ranking.
- 2. Relevance rules.** Define explicit rules for the two HR searches to reduce false positives and make automated screening auditable.
- 3. Human validation.** Validate rules against independent labels to confirm recruiter alignment and reduce manual review effort.
- 4. Semantic ranking.** Rank by search-intent similarity to improve candidate discovery despite differences in job-title wording.
- 5. Ranking quality.** Use NDCG@10 to confirm the strongest candidates appear first and reduce recruiter review time.
- 6. Feedback scalability.** Measure candidate movement and top-10 retention to personalize recommendations at scale without destabilizing strong shortlists.



# 1. Data Audit and Relevance Reference

The source contains 104 candidate rows but only 52 unique job titles. Ranking each unique title once prevents repeated text from receiving disproportionate weight. Fourteen titles repeat and the largest exact-title frequency is seven.



## Rule-based relevance

The query-specific reference is  $R = H(0.70 + 0.30I)$ . H is occupational relevance and I is intent alignment. Because intent is multiplied by occupational relevance, generic seeking-employment wording cannot become HR-relevant on intent alone. Clear employer solicitations receive zero candidate relevance.

| Validation measure               | Result        |
|----------------------------------|---------------|
| Exact ordinal agreement          | 39/52 (75.0%) |
| Quadratic weighted Cohen's kappa | 0.888         |
| Spearman correlation             | 0.871         |

## Vocabulary structure

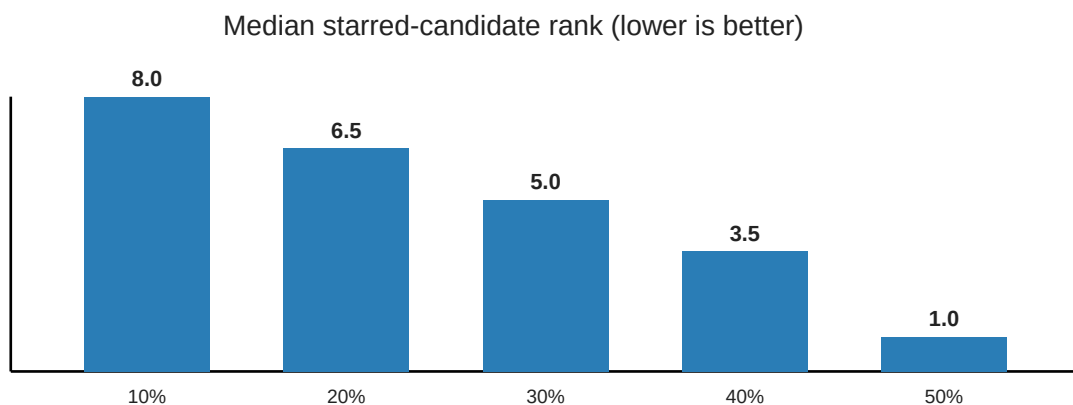
After deduplication, 31/52 titles explicitly contain Human Resources, 34/52 contain direct HR evidence, 12 contain aspiring, and 10 contain seeking. This supports an auditable rule layer plus semantic ranking.



### 3. Recruiter Feedback and Reranking

A starred candidate shifts the query as  $q_2 = \text{normalize}((1-w)q + w d^*)$ . The experiment evaluates 34 strong title/query scenarios at five feedback weights.

| Feedback | Median starred rank | Median top-10 retained |
|----------|---------------------|------------------------|
| 10%      | 8.0                 | 9/10                   |
| 20%      | 6.5                 | 9/10                   |
| 30%      | 5.0                 | 9/10                   |
| 40%      | 3.5                 | 8/10                   |
| 50%      | 1.0                 | 8/10                   |



#### Interpretation

No tested weight satisfies the former joint target of median starred rank  $\leq 2$  while retaining at least 9/10 of the original top ten. Model 40 therefore does **not** claim a universal optimum. At 30%, both representative starred candidates move 2 -> 1 while retaining 9/10 and 10/10 of the original top ten.

| Query / candidate  | Rank   | Top-10 retained | NDCG@10        |
|--------------------|--------|-----------------|----------------|
| Aspiring HR - ID 3 | 2 -> 1 | 9/10            | 0.948 -> 0.955 |
| Seeking HR - ID 99 | 2 -> 1 | 10/10           | 0.935 -> 0.940 |

## 4. Practical Recommendation and Limitations

Use the rule framework as an auditable relevance screen, Model 40 as the semantic ordering mechanism, and recruiter feedback as a controlled personalization layer. Avoid a universal cosine-similarity cutoff because similarity is query- and pool-dependent.

| Limitation         | Implication                                                                          |
|--------------------|--------------------------------------------------------------------------------------|
| Mean pooling       | Long titles can dilute important HR terms.                                           |
| Limited context    | Word2Vec cannot fully resolve directionality or sentence meaning.                    |
| Title-only feature | Skills and experience absent from the title cannot affect ranking.                   |
| Rule portability   | New occupations require new role/intent definitions and validation.                  |
| Feedback strength  | Higher weight moves starred candidates up faster but disrupts more of the shortlist. |

### Reproducibility

Notebook 01 runs from the included CSV files. Notebooks 02 and 03 require the official NLPL **40.zip** archive and stream only the vectors needed for the 52 titles and two queries. Official archive: <https://vectors.nlpl.eu/repository/20/40.zip>